

A Case Study of User Experience Design in a Disrupted Context: Design and Development of a Vital Signs Self-monitoring System

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ABSTRACT

We present a case study where we developed an interface for remote vital signs self-monitoring at a large-scale isolation facility for COVID-positive patients, under disrupted conditions. These conditions were: a lack of time, lack of access to end users, changing requirements, high risk of infection and supply chain limitations. We first describe the background of the development of the facility and the vital signs self-monitoring system. We use the 5 commonly prescribed activities of user experience design - Empathise, Define, Ideate, Prototype and Test- to describe how our work as user experience designers was affected by these disrupted conditions. Finally, we recommend a focus on Empathy, Prototyping and Communication for user experience practitioners and educators, whose training may be needed in similarly mission-critical, time-constrained circumstances.

CCS CONCEPTS

• **Human-centered computing** → **Interaction design process and methods.**

KEYWORDS

vital signs monitoring, health systems, design, disruption, emergency

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1 INTRODUCTION

User experience designers now work in many branches of private and public institutions. As the recognition of the role of interface design in high quality systems has grown, so has the demand grown for user-centered, experience-focused designers. During Singapore's early days of the response to the growing COVID-19 outbreak, a field hospital, referred to as community care facility (CCF), was established to isolate and house thousands of COVID-positive patients while they were waiting to recover [4]. This CCF was located at a convention center with 10 halls, capable of housing up to 10,000 patients. A key element of this field hospital was a vital signs monitoring system for patients to conduct their own monitoring. The two lead authors (Lim and Foong) supported the technology development work with interface design and developing patient education materials. A third author (Koh) acted as the clinical lead of the operations. All three authors were on site during the period of the case study. The final author, (Yeow), supported the study of the business case.

However, the conditions of work were vastly different from our usual practice. The development was rapid (5 days), yet it was also mission critical amidst changing environmental conditions. The hastily assembled medical, technology, nursing and operations personnel were working in conditions far beyond a 'normal' emergency. The situation was complicated by shifting clinical knowledge of the disease and supply chain disruptions. Upon preparing to write this case study, we were initially lost in describing the extreme circumstances we were under.

Organizational science, the study of organizational response, offers the term *disrupted* context from Hälgren, Rouleau and De Rond [8]. This oft-cited study consists of a metareview of 138 extreme context studies. They found that the work clusters meaningfully around *risky* contexts, *emergency* contexts and *disrupted* contexts.

In the simplest terms, a *risky* context is one where risk has been foreseen, and plans have been made to address these potential events. A HCI study in this space might contribute a system for firefighting in a virtual training environment. In contrast, during an *emergency* context, an event that is related to the organization's purpose actually occurs and requires urgent response. In HCI, this may be an evaluation study of a firefighters' incident command centre when in use. Finally, there is a class of extreme contexts

where the precipitating event is external to the organization and is not part of the organization's daily work. The work in this area studies the organization's response to the *disrupted* context. For example, HCI researchers might study the role of online communities in response to a natural disaster. A distinguishing element of disruptive contexts is that the "unique, unprecedented" event is external to the organization, and "often catches organizations off guard" [8, p. 135]. Another distinguishing feature is the emergence of *temporal organizations*, transient organizations consisting of personnel drawn from multiple sources who must coordinate to address the issues created by the disruption.

HCI research has extensively studied the behavior of people and information communication platforms within a disrupted context. The research most often covers the features social media use and citizen generated content, and the desirable feature sets when such applications are developed, e.g. [9, 14, 17]. However, there has been little work regarding the role of the designer and design methods within disrupted contexts. Our motivation is to contribute a case study on **how user experience designers who are deployed in disrupted contexts may support the design and development of ICTs**.

Our goal is to describe the design methods we used to support the delivery of a viable system even under disrupted conditions. In doing so, we also reflect on the elements of design training that remained robust despite the extenuating circumstances. We summarize the design approach in a disrupted context as E-P-C – to Empathize, Prototype and Communicate.

2 BACKGROUND: ESTABLISHMENT OF CCF@EXPO IN SINGAPORE

In 2020, the COVID-19 pandemic had taken the world by surprise with its unexpected clinical features that are unlike the previous coronavirus infections. The rapid spread of the infection, person-to-person transmissibility, and long infectious period of approximately 2 weeks necessitated the establishment of large-scale CCF, such as those built in India [2] and China [7].

During the initial period of the epidemic from February 2020 to April 2020 in Singapore, the number of COVID-19 cases rose rapidly and was predicted to keep rising (Table 1). Similar to China and India, our hospitals were at risk of running out of sufficient bed capacity for the increasing number of patients. The government quickly formed a multi-organization team to oversee the establishment of a CCF in Singapore at a 10-hall convention center called the Expo.

As this was a new facility, no dedicated staff for the facility were available. The facility had to draw manpower from existing health-care organizations to help with the operations. Three authors of this paper hold roles at the Telehealth Core, an eHealth research group, that also contributed its resources to the setup of the interactive system at the CCF. Table 1) details the relevant timeline events.

2.1 Environment of the CCF@Expo

The purpose of this CCF was to isolate COVID-positive patients who are clinically stable until they are no longer infectious. Approximately 80% of COVID-19 patients do not require hospital

admission [18]. However, these patients need to be monitored for acute oxygen desaturation or other rapid deterioration events [16].

The clinical authorities determined that at minimum, there should be 3 measurements per day at 8 hour intervals, of blood pressure, pulse rate, temperature and oxygen saturation. Furthermore, there was increasing scientific evidence that suggested that patients had a higher risk of respiratory complication at about 5 – 8 days from the onset of symptoms [16]. In normal hospital-based care, it falls to the nurses to monitor vital signs on a pre-determined schedule. However, at this CCF, the target nurse to patient ratio was 8-12 nurses per hall of 960 patients (at least 1: 80). Thus, it was not feasible for vital signs monitoring to be done by nurses.

In summary, due to the virus' ability to spread widely and rapidly within the population and a long recovery period, this outbreak had created an unprecedentedly low staff-to-patient ratio. Hence, there emerged, among other needs, a mission-critical need to use digital technology to help upscale monitoring capacity by having patient self-monitor at scale.

2.2 Target Users

The target users were all COVID-19 patients decanted to the CCF@Expo after an initial hospital-based review. In Singapore, a majority of the COVID-19 patients were migrant workers from the construction and building industry, many of them originating from South Asia, with a handful from China and other Asian countries [3]. Related research has documented the substantial barriers to healthcare in this population due to the language differences [5] between the health care officials who speak mostly English or Mandarin, and the patients who speak a large variety of languages and dialects from their countries of origin.

2.3 Requirements

When we began work at the CCF@Expo, the requirements for us as user experience designers were:

- (1) Support the design of a vital signs self-monitoring system where patients are guided to use 3 devices to enter readings into 6 fields on an iPad app for temperature, blood pressure (Systolic, Diastolic and pulse rate) and oxygen saturation (SpO2% and pulse rate), followed by 2 self-report measures of wellness and shortness of breath.
- (2) The patients should be able to monitor their vital signs at least 3 times a day.
- (3) Support the training of nurses to use the monitoring dashboard to review the patients' data.
- (4) High learnability for device guides and process of data entry for low English literacy users
- (5) High readability (visually scannable) for nurse dashboard to detect threshold alerts and monitoring compliance issues
- (6) Quick, mass deployment of the mission-critical system within 5 days.

3 METHOD

We used the case study method to document the experience of developing the interface. Besides the experience of the authors who were contributing their skillsets at the CCF, we present additional viewpoints from 7 interviews with key medical, nursing, operations

Table 1: Timeline of key events in the first 7 days of the establishment of the CCF@Expo, where the Ministry of Health (MOH), National Centre for Infectious Diseases (NCID), Ministry of Health Office of Healthcare Transformation (MOHT) and Integrated Health Information Systems (IHIS) initiated work on the Vital Signs Self-Monitoring System (VSSM). 4 April is the day after the national lockdown began.

Date (Day)	No. of Cases	Events in Singapore
4 April (Sat)	New=49 Total=1,049	• MOH's high-level requirements for a tele-monitoring system specified as: - Measure key vital signs, particularly (1) blood pressure, (2) heart rate, (3) blood oxygenation (SpO2) and (4) temperature 3 or 4 times a day; - Gather qualitative information from patients, such as whether or not the patient feels breathless; - Able to transmit, store and retrieve the data collected in a secure environment that complies with MOH's overall security requirements.
5 April (Sun)	New=75 Total=1,189	• Official announcement of large scale Community Care Facility at Expo (CCF@Expo) • Prototype set of Bluetooth-enabled monitors and VSSM patient-facing software passed to NCID to test. • Telehealth Core (THC) activated to support patient training for monitoring devices • IHIS started reaching out to vendors to procure tablets and vital signs monitoring devices.
6 April (Mon)	New=66 Total=1,375	• Urgent need for funding approval to procure tablets & devices because of large numbers required, competition from global markets and broken supply chains. • Application for budget for immediate set-up of system at CCF@Expo, targeted opening on 11 Apr (Saturday). • Bluetooth-connected devices assessed to be unfeasible at scale, so decision made that patients will measure their own vital signs and enter data into tablet. • MOH identifies critical limitation - patients MUST be able to do their own vital signs measurement, with sufficient infection control and without drawing down on nurse resources
7 April (Tues)	New=106 Total=1,481	• National University of Singapore's Medical Students' Society's Constructing Care Collaboration (CCC) group activated for translators for 4 languages • Telehealth Core (THC) begins redesign of community monitoring system interface to remove Bluetooth guide, toward manual data entry. • Final go ahead given for the use of IHIS' existing community monitoring systems at CCF@Expo
8 April (Wed)	New=142 Total=1,623	• 5am: Budget approved & IHIS informed to proceed with procurement of tablets & devices. • Medical, tech, operations teams convene at the CCF
9 April (Thurs)	New=287 Total=1,910	• Available blood pressure meters (60 sets) from IHIS' community care program brought to the CCF. • THC delivers first multi-lingual device manuals in visual/storyboard format, as well as dashboard manuals for medical staff • Clinical Team asks for Shock Index (SI) to be added to vital signs monitoring dashboard. • IHIS trains first 60 Woodlands Health Care staff to use dashboard
10 April (Fri)	New=198 Total=2,108	• MOH decides to open first halls (Halls 3 & 4) in CCF by 5pm (a day in advance). • 500 individual sets of tablets, BP/SpO2/thermometers & tablets (MVP 1.0) packed for distribution at patient registration • User manuals English, Chinese, Malay, Tamil and Bengali distributed in the CCF

and technology stakeholders over 2 weeks at the end of May 2020. We collaborated with medical students to conduct these interviews. Permission to conduct the interviews was obtained from the various organizations. The study was exempted from review by the university's ethics board as falling under the study of public service programs.

3.1 User Experience Design frameworks

There are many frameworks that describe how to apply design methods, particularly as it applies to interaction design. Variations exist across the user experience practice, with as few as 3 or as many as 9 intermediate steps. A discussion on the comparative value of these models is outside the scope of this case study. Arguably, the most popular variant is a 5 step process of consisting of

Empathize, Define, Ideate, Prototype and Test. Overall, the labels in this framework map reasonably well to theoretical work on design approaches, such as Lawson's How Designers Think [10] or Stolterman and Nelson's Design Way [13]. Hence, we use the descriptions of this model, as sourced from <http://interaction-design.org/> [1], to structure presentation of the work done in creating the vital signs self-monitoring system at the CCF@Expo.

4 OUTCOMES

4.1 Empathize

In this stage, the key activity is to "research your users' needs" in order to gain empathy to their particular needs. The process of interacting with end users and eliciting requirements helps designers

to “gain real insight into users and their needs.” The usual methods are interviews, participatory design or other ethnographically-informed methods.

4.1.1 Disruptive Context Limitations. The prescribed methods of elicitation and empathy development were severely limited by our time constraints and access constraints. Due to infection control measures, we were not permitted to enter the halls to have direct access to the patients.

Knowing this, we strove to get permission to, and then set up remote calls to patients identified by the health-care workers. However, in our efforts to do so, we saw that even drawing upon health-care workers to support these efforts represented a burden to them as they were already dealing with registration difficulties, training and rapid setup challenges.

Eventually, two video calls were arranged but were then limited by the language barrier and a severe shortage of translators. Volunteer translators for Bengali and Tamil were already fully occupied with communicating logistics or welfare material for the CCF and the migrant worker dormitories. As a result, these voice calls were conducted in simple English, leading to a limited ability to ask or understand complex issues such as queuing conditions, user challenges and system usability. Overall, direct interaction with end users was near impossible.

Given the lack of user contact, we turned our focus to ensuring that user training went smoothly. We first explored the use of training videos for vital signs monitoring. We spent time to develop video-based tutorials for Bluetooth-enabled devices, but delivery of said videos became difficult. The large number of occupants meant that internet bandwidth for video viewing could be severely compromised. Patient registration processes were still being worked out, which meant that phone numbers were not consistently available for sending video links. Finally, the decision to change to non-Bluetooth-enabled hardware quickly outdated the videos made just days earlier. In the end, we fell to developing paper guides for printing and distribution.

The use of continuous Bluetooth-enabled monitoring devices would have reduced the need to develop a tablet interface for the data entry, and to train users to enter the appropriate data. However, the development team at IHIS quickly discovered that re-tethering a lost Bluetooth connection meant choosing from potentially hundreds of available signals that will cause user confusion. Secondly, due to global supply chain disruptions, there were not enough Bluetooth-enabled devices to purchase.

Hence the tech team had to use a tablet-based data entry system with the traditional vital signs monitoring devices. In order to increase learnability for non-English speaking users, the interface and user guides had to incorporate strong graphical elements. Finally, in order to guide effectively, we made sure that all the graphics on paper and on the interface were coherent and look the same.

4.1.2 Coping Response. Without access to users, we turned to becoming user advocates instead. We used the following tools: personas and scenarios and cognitive walkthroughs. In doing so, we leveraged on our previous experience with other patients or with usability testing in general. Lim and Foong are both researchers and designers in their current role at the Telehealth Core, with more than 10 years of experience in the development and testing

of interactive systems. Lim in particular, leveraged his previous experiences as an ambulance medic to craft a user persona with low English and health literacy. We requested a set of monitoring equipment (blood pressure monitor, oral thermometer and pulse oximeter), and tested the self-monitoring process on ourselves, including the data entry process on the tablet.

4.1.3 Outcome. By doing this, we discovered the potential challenges of vital signs self-monitoring. We determined that a) the process takes a minimum of two minutes, and b) the order of measurements can impact the accuracy of the measurement. Critically, we saw that:

- (1) Forgetting to remove the blood pressure cuff could lead to an inaccurate oxygen saturation and pulse reading. Therefore, it was necessary to have the patients conduct the pulse oximeter measurement first before the blood pressure measurement.
- (2) When removed from the finger before data entry, the pulse oximeter display turns off, making it difficult to remember and enter the pulse and oxygen saturation readings. Hence, catering to right-handedness, the pulse oximeter had to be measured on the left hand in order to sustain the display while the right hand can be used to enter the data into the tablet (Fig. 1, right).

Once we had this information, we moved to inform the hardware and software setup personnel to anchor the monitors in the appropriate order on the telehealth kiosk (Fig 2, right). We also communicated the data on time to complete the monitoring. This was then used to estimate the number of kiosks needed for a hall of patients to complete monitoring within 1-2 hours (2 minutes, 960 patients, distributed over approximately 12 kiosks). We also to edited the patient education materials for this set up (Fig. 2, left).

In our case study interviews, the head of technology development reflected on the development of the education materials and kiosks. Among other learning points, he pointed out the value of the user experience input on both the hardware and software design process, placing particular emphasis on the “need for insight on patient behaviour” and the need to understand the particular characteristics of working with people with different technology and language expectations. He contrasted the attainment of the current project with another idea early in the process to send a hyperlink to the vital signs monitoring website via SMS. This idea was quickly discontinued due to language barriers that limit the ability to communicate the purpose of the link to the patients.

4.2 Define and Ideate

In the **definition** stage, the designers create materials to communicate the Users’ Needs and Problems as they have understood them. Designers may also synthesize the results of the user research phase and generate problem statement(s). In the **ideation** phase, designers generate different solutions based on the outcomes of the definition phase [1]. The goal is to broaden options and in so doing increase the likelihood of a better solution via stakeholder buy-in.

4.2.1 Disruptive Context Limitations. The events in this case study were taking place during the early days of the pandemic, where

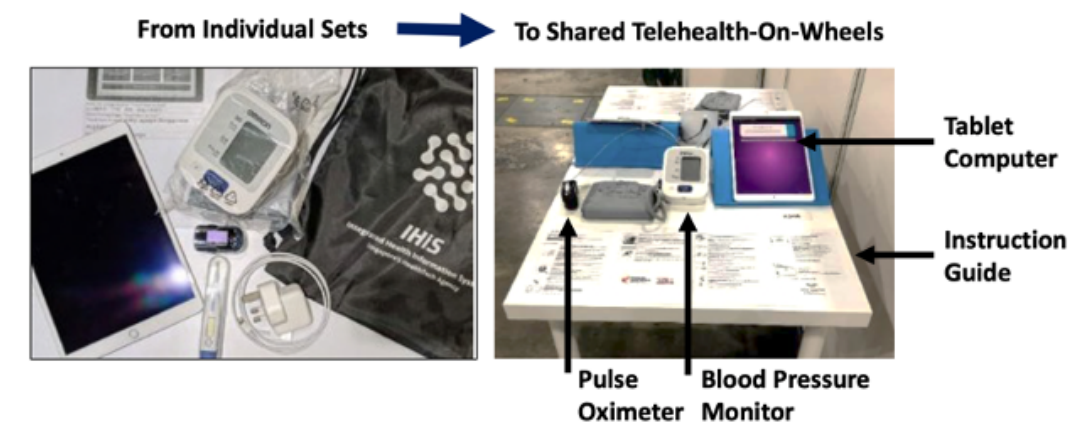


Figure 1: Evolution of vital-sign monitoring system used by patients in the CCF@Expo

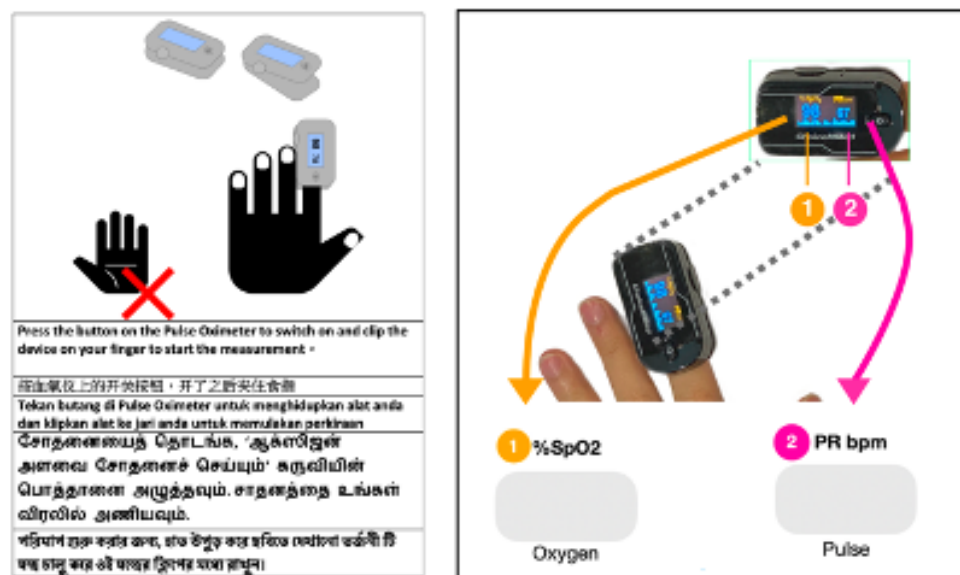


Figure 2: Visual elements in the paper guide to communicate measuring method (left), and in the tablet display to reduce the likelihood of users wrongly switching the oxygen saturation and pulse rate data.

disease characteristics were continually being discovered. These new developments were changing the care guidelines, which in turn limited our ability to stabilize system requirements. One such example was the growing recognition at the time that silent hypoxia (too little oxygen in the blood, without obvious signs of patient distress) was a key danger[6]. To ensure accurate detection for signs of hypoxia, we had to immediately revise the interface to increase emphasis on the correctness of the oxygen saturation (SpO₂) data entry into the system (Fig 2, right). Since there were two readings on the pulse oximeter display, there was a risk that users might accidentally switch the two readings. When switched, it could increase

the incidence of false positives (pulse rate mistaken as low oxygen saturation), or worse, false negatives (low oxygen saturation read as normal pulse rate). Another limitation was the rapid deployment needed. As mentioned above, case numbers were escalating, leading to a need to increase occupancy as quickly as possible. The extremely short development period (about 5 days) was further complicated by changing clinical requirements and hardware limitations. The global pandemic had impacted the supply chains. There were not enough tablets to be issued for every individual patient, so the development team had to increase capacity by moving from individual use systems to a kiosk-based system (Fig 1). This meant

that, in order to adapt the interface for multiple users, we would have to introduce language selection screens and user identification screens.

4.2.2 Coping Response. Due to the lack of time and the changing requirements, we could not have any clear definition or ideation stages. After being asked to design the interface, we used our understanding of the situation, and ongoing clarification with clinical and development staff to determine the requirements of the system. In essence, we moved straight to prototyping, as described in the next section.

4.3 Prototype and Communicate

During the prototyping phase of the 5-step framework, selected ideas go into lower-fidelity development. As mentioned above, we were forced to move immediately to prototyping due to lack of time and ongoing changes.

4.3.1 Outcomes. One common way that prototypes are understood is that they are a manifestation of design ideas. However, in academic design research texts, a second, less obvious use of prototypes is often surfaced – as a basis for communication. For example, Lim et al [12] discussed the use of prototypes to manifest design ideas, but also to filter and reduce the number of design ideas among stakeholders. Nelson and Stolterman [13] described how the envisioned design idea can support the conversation around the alignment of goals.

While our main purpose for creating prototypes was to manifest the design idea, we found that the manifestation thereof was the basis by which the multiple parties at the CCF could more quickly come to an alignment on what the interface can and should do. We created clickable prototypes using AdobeXD, then either discussed it with co-located personnel at the CCF, or sent a prototype out for discussion in our chat groups to gather feedback. Additionally, the prototypes enabled initial testing with nurses, who were also acting as proxy users for the patients. The nurses also reviewed the low-fidelity prototypes via the shared chatgroups or in person. Then they gathered feedback among themselves and passed it back to the developers who were co-located on the premises of the CCF. During the case study interviews, the head nurse reflected that this practice of sharing prototypes as they were created was productive because it created “easy access” and “unusual speed” of development, especially in comparison with her own experience with much slower systems development in hospitals.

4.4 Test/Evaluate

Although some form of testing is always present in the design process, this phase of the 5-step process formalizes it. This is usually through a testing protocol, using think-aloud, observational methods, quantitative measures or users’ self-report measures to carefully assess the system for usability and usefulness. The value of this step lies in the data gathered, and how it can be used to convince stakeholders towards a change, or to cement design decisions [1].

4.4.1 Disruptive Limitations . Again, due to the infection control measures and the time crunch, there was no opportunity to implement any kind of formalized testing. It was only about two weeks after the first version was released and some of the setup processes were settled, that we were finally able to reach out to nurses to get feedback on the patients’ use of the vital signs self-monitoring system. Similarly, IHIS took two weeks to invite a data analyst to collect and analyze compliance rates of vital signs monitoring.

4.4.2 Outcomes of Later Evaluation. This is not to say that evaluation is to be ignored. Eventually, we conducted an open-ended web survey with the nursing staff, asking how the patients were doing in their use of the vital signs self-monitoring systems, and how the nurses were working to ensure compliance. There were five respondents. The results are listed below.

- (1) Causes of non-compliance:
 - Patients forgot to do vital signs monitoring despite training received at registration.
 - Nurses found it difficult to communicate the need to do vitals sign monitoring, due to language barriers.
 - Non-registered patients were unable to login, therefore unable to log data.
- (2) Nurse strategies to improve vital signs monitoring compliance:
 - (re) Teach in person.
 - Remind in person.
 - Set reminder alarm on patients’ phone.
 - Ask their roommate to remind them.

Here, it seemed that the presence of the nurses was the key factor in driving the system usage and compliance rates. However, in our case interviews, a more complete view emerged. The head nurse reported that generally the patients were able to quickly learn after initial training at admission, particularly with the help of pictorial paper guides in hand, and on the kiosks to reinforce learning (Fig 1, right). The additional support supplied by the visual guides seemed to have helped in reducing the burden of training all patients. This, in turn, freed the nurses to spend time on pursuing a smaller number of patients who were not recording their vital signs.

5 LEARNING POINTS: EMPATHIZE, PROTOTYPE, COMMUNICATE

Upon reflection of our experience of user experience design in a disrupted context, we saw that we skipped the stages of problem definition, ideation and formal testing. Despite not including these key stages of a well-known UX design process, we were able to support the development of a usable vital signs monitoring system. We reflect here on our key lessons for designing in disruptive circumstances, in order to benefit future designers and educators of designers.

Firstly, despite the time crunch, we did not dismiss the need to empathize first. In this case study, we showed that our experience, and the additional time and effort of becoming user advocates led to productive changes in the interface. Hence, **empathy for the users, which lead to understanding the context of**

use and the limitations of use, is critical even in a disrupted context.

In order to become adequate user advocates, we drew on our previous experiences of extensive testing with a wide variety of user groups. Similarly, making rapid decisions that accommodate both the design requirements and circumstances required *design judgement*. Nelson and Stolterman, in their discussion of processes that contribute to design success, describe judgement as being "based on accessing knowledge that is generated in the particularity or uniqueness of a situation; *knowledge that is inseparable from the knower* and is only revealed through the actions of the knower." [13, p. 181]. Hence, the key takeaway for educators and practitioners is that an investment in gaining personal knowledge via ongoing interactions with diverse users will advance their ability to use design judgement in disrupted contexts.

Secondly, **when working in extreme contexts, it is more important to attain the bare minimum of functionality, than to achieve design quality.** Tohidi et al's [15] oft-cited paper on getting the design right suggests that ideation generates more elaborate and more honest prototype feedback from test users. When we skipped that step, it is very likely that we lost out on generating better ideas. However, the need to monitor patients and avert potentially fatal health crises took precedence over creating a high quality interface at the first pass. In this, we were fortunate that the team leaders at the CCF identified and communicated their key, urgent business goals as stated in Table 1. By doing so, the various personnel on site, including the designers, were able to prioritize the most important tasks.

Thirdly, we found it was critical to **"open" the design process to invite conversation and communication.** We were unable to test the system rigorously enough from a usability perspective. Even though there were no testing efforts until weeks after the initial implementation, we saw that gathering usage data is still valuable for later iterations. Thus, we do not recommend eliminating testing in a disruptive context. Rather, the key lesson is that when forced to forgo testing, the use of a prototype to gather feedback and communicate ideas from stakeholders was a viable interim measure to improve the robustness of the design. As described above, the interface design of the oxygen saturation input screen (Fig. 2) evolved from discussions with the hardware, clinical and user experience personnel. In overcoming disruptive contexts, a focus on communication to align multiple stakeholders can contribute to success [11]. User experience designers can support this effort with rapid, low-fidelity prototypes that act as a cognitive scaffold for moving the design forward.

6 CONCLUSION

We contribute a case study of using user experience design methods in a disrupted context where designers may experience extreme constraints in terms of time, hardware and connectivity options, lacking access to users and the use environment, as well as rapidly changing requirements. We compared our approach against a popular design framework and conclude that in disruptive contexts, it is critical to retain the stages of empathy and prototyping, and to use the prototype to facilitate communication among multiple stakeholders.

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REFERENCES

- [1] [n.d.]. What is Design Thinking? <https://www.interaction-design.org/literature/topics/design-thinking>
- [2] 2020. Indian capital readies vast quarantine centre as coronavirus cases mount | Reuters. <https://www.reuters.com/article/us-health-coronavirus-india-idUSKBN23W1Z9>
- [3] 2020. Majority of recent Covid-19 cases among foreign workers are those in quarantine facilities and have old infections. *The Straits Times* (Aug. 2020). <https://www.straitstimes.com/singapore/majority-of-recent-covid-19-cases-among-foreign-workers-are-old-infections-from-quarantine>
- [4] 2020. Singapore Expo being prepared to house COVID-19 patients who have recovered but may still be infectious: Gan Kim Yong. *CNA* (April 2020). <https://www.channelnewsasia.com/news/singapore/covid19-singapore-expo-community-care-facility-12611318>
- [5] Jw Ang, Cj Koh, Bwb Chua, S Narayanaswamy, L Wijaya, Lg Chan, Li Soh, Wl Goh, and S Vasoo. 2020. Are migrant workers in Singapore receiving adequate healthcare? A survey of doctors working in public tertiary healthcare institutions. *Singapore Medical Journal* 61, 10 (Oct. 2020), 540–547. <https://doi.org/10.11622/smedj.2019101>
- [6] Atanu Chandra, Uddalak Chakraborty, Jyotirmoy Pal, and Parthasarathi Kar-makar. 2020. Silent hypoxia: a frequently overlooked clinical entity in patients with COVID-19. *BMJ Case Reports* CP 13, 9 (Sept. 2020), e237207. <https://doi.org/10.1136/bcr-2020-237207> Publisher: BMJ Specialist Journals Section: Reminder of important clinical lesson.
- [7] Simiao Chen, Zongjiu Zhang, Juntao Yang, Jian Wang, Xiaohui Zhai, Till Bärnighausen, and Chen Wang. 2020. Fangcang shelter hospitals: a novel concept for responding to public health emergencies. *The Lancet* 395, 10232 (April 2020), 1305–1314. [https://doi.org/10.1016/S0140-6736\(20\)30744-3](https://doi.org/10.1016/S0140-6736(20)30744-3) Publisher: Elsevier.
- [8] Markus Höllgren, Linda Rouleau, and Mark de Rond. 2018. A Matter of Life or Death: How Extreme Context Research Matters for Management and Organization Studies. *Academy of Management Annals* 12, 1 (Jan. 2018), 111–153. <https://doi.org/10.5465/annals.2016.0017>
- [9] Marc-André Kaufhold, Christian Reuter, Christoph Amelunxen, and Massimo Cristaldi. 2018. 112. Social: Design and Evaluation of a Mobile Crisis App for Bidirectional Communication between Emergency Services and Citizens. 17.
- [10] Bryan Lawson. 2006. *How designers think: the design process demystified* (4. ed ed.). Elsevier/Architectural Press, Amsterdam. OCLC: 254426229.
- [11] Dorothy E. Leidner, Gary Pan, and Shan L. Pan. 2009. The role of IT in crisis response: Lessons from the SARS and Asian Tsunami disasters. *The Journal of Strategic Information Systems* 18, 2 (June 2009), 80–99. <https://doi.org/10.1016/j.jsis.2009.05.001>
- [12] Youn-Kyung Lim, Erik Stolterman, and Josh Tenenbergs. 2008. The anatomy of prototypes: Prototypes as filters, prototypes as manifestations of design ideas. *ACM Transactions on Computer-Human Interaction* 15, 2 (July 2008), 7:1–7:27. <https://doi.org/10.1145/1375761.1375762>
- [13] Harold G. Nelson and Erik Stolterman. 2012. *The design way: intentional change in an unpredictable world* (second edition ed.). MIT Press, Cambridge, MA. OCLC: 811124985.
- [14] Christian Reuter and Marc-André Kaufhold. 2018. Fifteen years of social media in emergencies: A retrospective review and future directions for crisis Informatics. *Journal of Contingencies and Crisis Management* 26, 1 (March 2018), 41–57. <https://doi.org/10.1111/1468-5973.12196>
- [15] Maryam Tohidi, William Buxton, Ronald Baecker, and Abigail Sellen. 2006. Getting the right design and the design right. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems (CHI '06)*. Association for Computing Machinery, New York, NY, USA, 1243–1252. <https://doi.org/10.1145/1124772.1124960>
- [16] Dawei Wang, Bo Hu, Chang Hu, Fangfang Zhu, Xing Liu, Jing Zhang, Binbin Wang, Hui Xiang, Zhenshun Cheng, Yong Xiong, Yan Zhao, Yirong Li, Xinghuan Wang, and Zhiyong Peng. 2020. Clinical Characteristics of 138 Hospitalized Patients With 2019 Novel Coronavirus-Infected Pneumonia in Wuhan, China. *JAMA* 323, 11 (March 2020), 1061. <https://doi.org/10.1001/jama.2020.1585>
- [17] Eric T White. 2014. The Application of Social Media in Disasters.
- [18] WHO. 2020. Media Statement: Knowing the risks for COVID-19. <https://www.who.int/indonesia/news/detail/08-03-2020-knowing-the-risk-for-covid-19>